

Progress Test 3.C — Thermo-S1

3.C · Equations of state of real gases

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The co-volume of carbon dioxide is $b = 4.27 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}$. Express it in $\text{cm}^3 \text{ mol}^{-1}$.

[illegible]

Exercise 2. The Van der Waals model

(5 marks)

Write the Van der Waals equation of state for n moles. Explain the physical meaning of the term involving b and of the term involving a . What does the equation reduce to when $a \rightarrow 0$ and $b \rightarrow 0$?

Marking

1	
0	1

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. The Amagat diagram

(4 marks)

In the Amagat diagram, when $p \rightarrow 0$ the compressibility factor Z tends to:

- ☐ 1 ☐ 0 ☐ $+\infty$ ☐ -1

At moderate pressure, $Z < 1$ for most gases because:

- ☐ attractive interactions dominate
 - ☐ the molecules' size dominates
 - ☐ the gas is ideal
 - ☐ the temperature is too high

The Boyle temperature T_B is the temperature at which:

- ☐ $Z \approx 1$ over a wide range of p
 - ☐ the gas liquefies
 - ☐ $Z = 0$
 - ☐ $a = 0$

According to Tab. 7, the gas with the strongest intermolecular attractions, and the highest critical temperature, is:

- ☐
- He
- ☐
- H
- ₂
- ☐
- N
- ₂
- ☐
- H
- ₂
- O

Exercise 4. Carbon dioxide under pressure

(4 marks)

One mole of CO_2 occupies the volume V at temperature T . Compute (a) the pressure p_{id} predicted by the ideal gas law; (b) the pressure p predicted by the Van der Waals equation; (c) the corresponding compressibility factor Z . Interpret.

$$\begin{array}{llll} n = 1.0 \text{ mol} & V = 0.50 \text{ L} & T = 300 \text{ K} & RT \approx 2.5 \times 10^3 \text{ J mol}^{-1} \\ a = 0.36 \text{ Pa m}^6 \text{ mol}^{-2} & b = 4.0 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1} & \text{(rounded values for CO}_2\text{)} & \end{array}$$

Marking

1

01

2

01

3

01

4

01

— End of the test —